TECHNICAL DATASHEET: LITHIUM BATTERY PACK

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1. PRODUCT IDENTIFICATION & ELECTRICAL CHARACTERISTICS

Parameter	Specification Value	Notes / Standards
Model Name / Number	**VLT-2026-G1**	Part Number: P/N-VLT-2026-X1
Battery Chemistry	Lithium Iron Phosphate (LFP / LiFePO4)	High safety, long cycle life
Nominal Capacity	**150 Ah**	Measured at 0.2C discharge rate
Nominal Voltage	**48 V**	Operating Voltage range: 40V to 54.8V
Cut-off Voltage	40 V	Lower safety threshold
Max Charge Current	75 A	Recommended charge current: 50A
Max Discharge Current	150 A	Peak discharge current: 200A (30s)
Energy Density	145 Wh/kg	Gravimetric energy density
Internal Resistance	12 mΩ	AC Impedance @ 1kHz

2. PHYSICAL & PERFORMANCE SPECIFICATIONS

Parameter	Specification Value	Notes / Details
Net Weight	**49.8 kg**	Total package mass
Dimensions	**482 x 350 x 176 mm**	Length x Width x Height
Terminal Type	M8 Terminal	Threaded copper terminals
IP Rating	**IP65**	Dust-proof and water-resistant casing
Expected Cycle Life	**4000 cycles**	At 80% DOD, 25°C ambient
Calendar Life	10 years	Operational lifespan
Expected Lifetime	**4000 cycles**	Expected Lifetime Cycles
Operating Temperature	-20°C to 60°C	Charge: 0~45°C; Discharge: -20~60°C
Storage Temperature	-10°C to 45°C	Recommended storage @ 50% SOC
Self-Discharge Rate	2.5%/month	Measured at 25°C

3. MANUFACTURER & GLOBAL COMPLIANCE PARTNERS

Entity Role	Name & Details	Contact & Address Info
Manufacturer	**VoltPower Technologies Ltd.**	Email: support@voltpower.in Address: Sector 62, Industrial Area, Noida, India
EU Authorized Representative	**EuroCompliance Representative GmbH**	Email: eu.rep@eurocompliance.de Address: Kaiserstrasse 42, 60311 Frankfurt, Germany
Country of Origin	**India**	Factory location: Noida, Uttar Pradesh, India

4. CERTIFICATIONS & REGULATORY COMPLIANCE

This LFP Battery Pack is fully certified and compliant with the following global standards:
• **UN38.3**: Transport safety certified (Lithium battery testing passed)
• **CE Marking**: Fully compliant with EU Directive (CE Mark applied)
• **Notified Body**: Certified by Notified Body Number 0123
• **IEC 62133**: Safety requirements for portable sealed secondary cells (IEC-62133 Certified)
• **BIS Certified**: Bureau of Indian Standards registration (**BIS Number: R-41123456**)
• **EPR Registered**: Extended Producer Responsibility (**EPR Number: EPR-BAT-2026-NOIDA**)
• **HSN Code**: Tariff Classification **85076000** (Lithium-ion accumulators)
• **RoHS & REACH**: 100% compliant, lead-free and mercury-free construction.
• **Warranty**: Covered by a **60 months** limited warranty from the date of manufacture.

5. ENVIRONMENTAL DECLARATIONS & MATERIAL COMPOSITION

• **Carbon Footprint**: **42.5 kg CO2e** per battery pack (Performance Class: **A**)
• **Removable & Replaceable**: The battery pack casing is designed for easy disassembly, allowing cell modules and BMS to be removed and replaced for repair or recycling.
• **Hazardous Substances Declaration**: Lead, mercury, and cadmium content is below thresholds. Hazardous substances declaration: 'Lead: absent, Mercury: absent, Cadmium: absent. Safe for consumer use.'

Material / Active Ingredient	Percentage (%)	Recycled Content (%)
Lithium (Li)	8.5 %	10.0 % (Recycled Lithium)
Cobalt (Co)	0.0 % (Cobalt-free LFP)	0.0 %
Nickel (Ni)	0.0 % (Nickel-free LFP)	0.0 %
Manganese (Mn)	0.0 %	0.0 %
Graphite (C)	14.5 %	15.0 % (Recycled Graphite)
Lead (Pb)	0.0 %	0.0 %

Recycled Content Summary: Total recycled content is 12.5%. Recycled Cobalt: 0%, Recycled Lithium: 10%, Recycled Nickel: 0%, Recycled Lead: 0%.